

The Foundations: Logic and Proofs

Chapter 1, Part I: Propositional Logic

With Question/Answer Animations

Chapter Summary

- Propositional Logic
 - The Language of Propositions
 - Applications
 - Logical Equivalences
- Predicate Logic
 - The Language of Quantifiers
 - Logical Equivalences
 - Nested Quantifiers
- Proofs
 - Rules of Inference
 - Proof Methods
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Propositional Logic Summary

- The Language of Propositions
 - Connectives
 - Truth Values
 - Truth Tables
- Applications
 - Translating English Sentences
 - System Specifications
 - Logic Puzzles
 - Logic Circuits
- Logical Equivalences
 - Important Equivalences
 - Showing Equivalence
 - Satisfiability

Propositional Logic

Section 1.1

Section Summary

- Propositions
- Connectives
 - Negation
 - Conjunction
 - Disjunction
 - Implication; contrapositive, inverse, converse
 - Biconditional
- Truth Tables

Propositions

- A *proposition* is a declarative sentence that is either true or false.
- Examples of propositions:
 - a) The Moon is made of green cheese.
 - b) Trenton is the capital of New Jersey.
 - c) Toronto is the capital of Canada.
 - d) $1 + 0 = 1$
 - e) $0 + 0 = 2$
- Examples that are not propositions.
 - a) Sit down!
 - b) What time is it?
 - c) $x + 1 = 2$
 - d) $x + y = z$

Propositional Logic

- Constructing Propositions
 - Propositional Variables: $p, q, r, s, \dots$
 - The proposition that is always true is denoted by **T** and the proposition that is always false is denoted by **F**.
 - Compound Propositions; constructed from logical connectives and other propositions
 - Negation $\neg$
 - Conjunction $\wedge$
 - Disjunction $\vee$
 - Implication $\rightarrow$
 - Biconditional $\leftrightarrow$

Propositional Logic

“Elephants are bigger than mice.”

Is this a statement? Yes

Is this a proposition? Yes

What is the truth value of the proposition? True

“ $520 < 111$ ”

Is this a statement? Yes

Is this a proposition? Yes

What is the truth value of the proposition? False

“ $y > 5$ ”

Is this a statement? Yes

Is this a proposition? No

Compound Propositions: Negation

- The *negation* of a proposition p is denoted by $\neg p$ and has this truth table:

p	$\neg p$
T	F
F	T

- **Example:** If p denotes “The earth is round.”, then $\neg p$ denotes “It is not the case that the earth is round,” or more simply “The earth is not round.”

Conjunction

- The *conjunction* of propositions p and q is denoted by $p \wedge q$ and has this truth table:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

- **Example:** If p denotes “I am at home.” and q denotes “It is raining.” then $p \wedge q$ denotes “I am at home and it is raining.”

Disjunction

- The *disjunction* of propositions p and q is denoted by $p \vee q$ and has this truth table:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

- **Example:** If p denotes “I am at home.” and q denotes “It is raining.” then $p \vee q$ denotes “I am at home or it is raining.”

The Connective Or in English

- In English “or” has two distinct meanings.
 - “Inclusive Or” - In the sentence “Students who have taken CS202 or Math120 may take this class,” we assume that students need to have taken one of the prerequisites, but may have taken both. This is the meaning of disjunction. For $p \vee q$ to be true, either one or both of p and q must be true.
 - “Exclusive Or” - When reading the sentence “Soup or salad comes with this entrée,” we do not expect to be able to get both soup and salad. This is the meaning of Exclusive Or (Xor). In $p \oplus q$, one of p and q must be true, but not both. The truth table for $\oplus$ is:

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

The Connective Or in English

- Check

$$p \oplus q \equiv (\neg p \wedge q) \vee (p \wedge \neg q)$$

Tonight I will stay home or go out to a movie.”

- p = “I will earn an A in this course,”
- q = “I will drop this course,”
- $p \oplus q$ = “I will either earn an A in this course, or I will drop it (but not both!)”

Implication

- If p and q are propositions, then $p \rightarrow q$ is a *conditional statement* or *implication* which is read as “if p , then q ” and has this truth table:

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

- **Example:** If p denotes “I am at home.” and q denotes “It is raining.” then $p \rightarrow q$ denotes “If I am at home then it is raining.”
- In $p \rightarrow q$, p is the *hypothesis* (*antecedent* or *premise*) and q is the *conclusion* (or *consequence*).

Understanding Implication

- In $p \rightarrow q$ there does not need to be any connection between the antecedent or the consequent. The “meaning” of $p \rightarrow q$ depends only on the truth values of p and q .
- These implications are perfectly fine, but would not be used in ordinary English.
 - “If the moon is made of green cheese, then I have more money than Bill Gates.”
 - “If the moon is made of green cheese then I’m on welfare.”
 - “If $1 + 1 = 3$, then your grandma wears combat boots.”

Understanding Implication (cont)

- One way to view the logical conditional is to think of an obligation or contract.
 - “If I am elected, then I will lower taxes.”
 - “If you get 100% on the final, then you will get an A.”
- If the politician is elected and does not lower taxes, then the voters can say that he or she has broken the campaign pledge. Something similar holds for the professor. This corresponds to the case where p is true and q is false.

Understanding Implication (cont)

- “If this lecture ever ends, then the sun will rise tomorrow.” True or *False*?
- “If Tuesday is a day of the week, then I am a penguin.” *True* or False?
- “If $1+1=6$, then Bush is president.”
True or *False*?
- “If the moon is made of green cheese, then I am richer than Bill Gates.” True or *False*?

Different Ways of Expressing $p \rightarrow q$

- if p , then q
- if p , q
- q unless $\neg p$
- q if p
- q whenever p
- q follows from p
- p implies q
- p only if q
- q when p
- p is sufficient for q
- q is necessary for p
- a necessary condition for p is q
- a sufficient condition for q is p

Different Ways of Expressing $p \rightarrow q$

p is the proposition "Ahmed learns discrete mathematics"

q is the proposition "Ahmed will find a good job"

The $p \rightarrow q$ is proposition

"if Ahmed learns discrete mathematics, then he will find a good job"

This proposition

Is false when p is true and q is false

Otherwise it is true

Converse, Contrapositive, and Inverse

- From $p \rightarrow q$ we can form new conditional statements .
 - $q \rightarrow p$ is the **converse** of $p \rightarrow q$
 - $\neg q \rightarrow \neg p$ is the **contrapositive** of $p \rightarrow q$
 - $\neg p \rightarrow \neg q$ is the **inverse** of $p \rightarrow q$

Example: Find the converse, inverse, and contrapositive of “It raining is a sufficient condition for my not going to town.”

Solution:

converse: If I do not go to town, then it is raining.

inverse: If it is not raining, then I will go to town.

contrapositive: If I go to town, then it is not raining.

Converse, Contrapositive, and Inverse

The conditional statement (implication) $p \rightarrow q$

"if it is raining, then the home team wins"

$\neg q \rightarrow \neg p$ is called **contrapositive** of $p \rightarrow q$

"if the home team does not win, then it is not raining"

The proposition $q \rightarrow p$ is called **converse** of $p \rightarrow q$

"if the home team wins, then it is raining"

$\neg p \rightarrow \neg q$ is called **inverse** of $p \rightarrow q$

"if it is not raining, then the home team does not win"

Converse, Contrapositive, and Inverse

- *Contrapositives:* $p \rightarrow q \equiv \neg q \rightarrow \neg p$?

Ex. “If it is noon, then I am hungry.”

“If I am not hungry, then it is not noon.”

YES

- *Converses:* $p \rightarrow q \equiv q \rightarrow p$?

Ex. “If it is noon, then I am hungry.”

“If I am hungry, then it is noon.”

yes

- *Inverses:* $p \rightarrow q \equiv \neg p \rightarrow \neg q$?

Ex. “If it is noon, then I am hungry.”

“If it is not noon, then I am not hungry”

yes

Converse, Contrapositive, and Inverse

- Proving the equivalence of $p \rightarrow q$ and its **contrapositive** using truth tables:

p	q	$\neg q$	$\neg p$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	T	F	F	F
F	T	F	T	T	T
F	F	T	T	T	T

Biconditional

- If p and q are propositions, then we can form the *biconditional* proposition $p \leftrightarrow q$, read as “ p if and only if q .” The biconditional $p \leftrightarrow q$ denotes the proposition with this truth table:

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

- If p denotes “I am at home.” and q denotes “It is raining.” then $p \leftrightarrow q$ denotes “I am at home if and only if it is raining.”

Expressing the Biconditional

- Some alternative ways “ p if and only if q ” is expressed in English:
 - p is necessary and sufficient for q
 - if p then q , and conversely
 - p iff q

Expressing the Biconditional

The proposition $p \leftrightarrow q$ has the same truth value as $p \rightarrow q \wedge q \rightarrow p$

p	q	$p \rightarrow q$	$q \rightarrow p$	$p \rightarrow q \wedge q \rightarrow p$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	T	F	F	F
F	F	T	T	T	T

Check $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

Truth Tables For Compound Propositions

- Construction of a truth table:
- Rows
 - Need a row for every possible combination of values for the atomic propositions.
- Columns
 - Need a column for the compound proposition (usually at far right)
 - Need a column for the truth value of each expression that occurs in the compound proposition as it is built up.
 - This includes the atomic propositions

Example Truth Table

- Construct a truth table for $p \vee q \rightarrow \neg r$

p	q	r	$\neg r$	$p \vee q$	$p \vee q \rightarrow \neg r$
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	F	T

Equivalent Propositions

- Two propositions are **equivalent** if they always have the same truth value.
- Example:** Show using a truth table that the biconditional is equivalent to the contrapositive.

Solution:

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg q \rightarrow \neg p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

Using a Truth Table to Show Non-Equivalence

Example: Show using truth tables that neither the converse nor inverse of an implication are not equivalent to the implication.

Solution:

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$\neg p \rightarrow \neg q$	$q \rightarrow p$
T	T	F	F	T	T	T
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	T	T	T

Problem

- How many rows are there in a truth table with n propositional variables?

Solution: 2^n We will see how to do this in Chapter 6.

- Note that this means that with n propositional variables, we can construct 2^n distinct (i.e., not equivalent) propositions.

Precedence of Logical Operators

Operator	Precedence
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

$p \vee q \rightarrow \neg r$ is equivalent to $(p \vee q) \rightarrow \neg r$
If the intended meaning is $p \vee (q \rightarrow \neg r)$
then parentheses must be used.

Applications of Propositional Logic

Section 1.2

Applications of Propositional Logic: Summary

- Translating English to Propositional Logic
- System Specifications
- Boolean Searching
- Logic Puzzles

Translating English Sentences

- Steps to convert an English sentence to a statement in propositional logic
 - Identify atomic propositions and represent using propositional variables.
 - Determine appropriate logical connectives
- “If I go to Harry’s or to the country, I will not go shopping.”
 - p : I go to Harry’s
 - q : I go to the country.
 - r : I will go shopping.

If p or q then not r .

$$(p \vee q) \rightarrow \neg r$$

Example

Problem: Translate the following sentence into propositional logic:

“You can access the Internet from campus only if you are a computer science major or you are not a freshman.”

One Solution: Let a , c , and f represent respectively “You can access the internet from campus,” “You are a computer science major,” and “You are a freshman.”

$$a \rightarrow (c \vee \neg f)$$

System Specifications

- System and Software engineers take requirements in English and express them in a precise specification language based on logic.

Example: Express in propositional logic:

“The automated reply cannot be sent when the file system is full”

Solution: One possible solution: Let r denote “The automated reply can be sent” and f denote “The file system is full.”

Therefore, we can translate this as $\neg r$ when f

$$f \rightarrow \neg r$$

Consistent System Specifications

Definition: A list of propositions is *consistent* if it is possible to assign truth values to the proposition variables so that each proposition is true.

Exercise: Are these specifications consistent?

- “The diagnostic message is stored in the buffer or it is retransmitted.”
- “The diagnostic message is not stored in the buffer.”
- “If the diagnostic message is stored in the buffer, then it is retransmitted.”

Solution: Let p denote “The diagnostic message is not stored in the buffer.” Let q denote “The diagnostic message is retransmitted” The specification can be written as: $p \vee q, \neg p, p \rightarrow q$. When p is false and q is true all three statements are true. So the specification is consistent.

- What if “The diagnostic message is not retransmitted is added.”

Solution: Now we are adding $\neg q$ and there is no satisfying assignment. So the specification is not consistent.

Consistent System Specifications

p	q	$\neg p$	$p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	T	F
F	T	T	T	T
F	F	T	F	T

system
specifications are
consistent

p	q	$\neg p$	$p \vee q$	$p \rightarrow q$	$\neg q$
T	T	F	T	T	F
T	F	F	T	F	T
F	T	T	T	T	F
F	F	T	F	T	T

system specifications
are inconsistent

Logic Puzzles



Raymond
Smullyan
(Born 1919)

- An island has two kinds of inhabitants, *knight*s, who always tell the truth, and *knave*s, who always lie.
- You go to the island and meet A and B.
 - A says “B is a knight.”
 - B says “The two of us are of opposite types.”

Example: What are the types of A and B?

Solution: Let p and q be the statements that A is a knight and B is a knight, respectively. So, then $\neg p$ represents the proposition that A is a knave and $\neg q$ that B is a knave.

- If A is a knight, then p is true. Since knights tell the truth, q must also be true. Then $(p \wedge \neg q) \vee (\neg p \wedge q)$ would have to be true, but it is not. So, A is not a knight and therefore $\neg p$ must be true.
- If A is a knave, then B must not be a knight since knaves always lie. So, then both $\neg p$ and $\neg q$ hold since both are knaves.

Propositional Equivalences

Section 1.3

Section Summary

- Tautologies, Contradictions, and Contingencies.
- Logical Equivalence
 - Important Logical Equivalences
 - Showing Logical Equivalence

Tautologies, Contradictions, and Contingencies

- A tautology is a proposition which is always true.
 - Example: $p \vee \neg p$
- A *contradiction* is a proposition which is always false.
 - Example: $p \wedge \neg p$
- A *contingency* is a proposition which is neither a tautology nor a contradiction, such as p

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Logically Equivalent

- Two compound propositions p and q are logically equivalent if $p \leftrightarrow q$ is a tautology.
- We write this as $p \leftrightarrow q$ or as $p \equiv q$ where p and q are compound propositions.
- Two compound propositions p and q are equivalent if and only if the columns in a truth table giving their truth values agree.
- This truth table show $\neg p \vee q$ is equivalent to $p \rightarrow q$.

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$



Augustus De Morgan

1806-1871

This truth table shows that De Morgan's Second Law holds.

p	q	$\neg p$	$\neg q$	$(p \vee q)$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

Key Logical Equivalences

- Identity Laws: $p \wedge T \equiv p$, $p \vee F \equiv p$
- Domination Laws: $p \vee T \equiv T$, $p \wedge F \equiv F$
- Idempotent laws: $p \vee p \equiv p$, $p \wedge p \equiv p$
- Double Negation Law: $\neg(\neg p) \equiv p$
- Negation Laws: $p \vee \neg p \equiv T$, $p \wedge \neg p \equiv F$

Key Logical Equivalences (*cont*)

- Commutative Laws: $p \vee q \equiv q \vee p$, $p \wedge q \equiv q \wedge p$
- Associative Laws: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
 $(p \vee q) \vee r \equiv p \vee (q \vee r)$
- Distributive Laws: $(p \vee (q \wedge r)) \equiv (p \vee q) \wedge (p \vee r)$
 $(p \wedge (q \vee r)) \equiv (p \wedge q) \vee (p \wedge r)$
- Absorption Laws: $p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$

More Logical Equivalences

TABLE 7 Logical Equivalences Involving Conditional Statements.

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

TABLE 8 Logical Equivalences Involving Biconditional Statements.

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

More Logical Equivalences

Using equivalences, we can *define* operators in terms of other operators.

Implication: $p \rightarrow q \equiv \neg p \vee q$

Biconditional: $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

$$p \leftrightarrow q \equiv \neg(p \oplus q)$$

Exclusive or: $p \oplus q \equiv (p \vee q) \wedge \neg(p \wedge q)$

$$p \oplus q \equiv (p \wedge \neg q) \vee (q \wedge \neg p)$$

Constructing New Logical Equivalences

- We can show that two expressions are logically equivalent by developing a series of logically equivalent statements.
- To prove that $A \equiv B$ we produce a series of equivalences beginning with A and ending with B .

$$\begin{array}{c} A \equiv A_1 \\ \vdots \\ A_n \equiv B \end{array}$$

- Keep in mind that whenever a proposition (represented by a propositional variable) occurs in the equivalences listed earlier, it may be replaced by an arbitrarily complex compound proposition.

Equivalence Proofs

Example: Show that $\neg(p \vee (\neg p \wedge q))$
is logically equivalent to $\neg p \wedge \neg q$

Solution:

$\neg(p \vee (\neg p \wedge q))$	$\equiv$	$\neg p \wedge \neg(\neg p \wedge q)$	by the second De Morgan law
	$\equiv$	$\neg p \wedge [\neg(\neg p) \vee \neg q]$	by the first De Morgan law
	$\equiv$	$\neg p \wedge (p \vee \neg q)$	by the double negation law
	$\equiv$	$(\neg p \wedge p) \vee (\neg p \wedge \neg q)$	by the second distributive law
	$\equiv$	$F \vee (\neg p \wedge \neg q)$	because $\neg p \wedge p \equiv F$
	$\equiv$	$(\neg p \wedge \neg q) \vee F$	by the commutative law for disjunction
	$\equiv$	$(\neg p \wedge \neg q)$	by the identity law for F

Equivalence Proofs

Example: Show that $(p \wedge q) \rightarrow (p \vee q)$
is a tautology.

Solution:

$(p \wedge q) \rightarrow (p \vee q)$	$\equiv$	$\neg(p \wedge q) \vee (p \vee q)$	by truth table for $\rightarrow$
	$\equiv$	$(\neg p \vee \neg q) \vee (p \vee q)$	by the first De Morgan law
	$\equiv$	$(\neg p \vee p) \vee (\neg p \vee \neg q)$	by associative and commutative laws
			laws for disjunction
	$\equiv$	$T \vee T$	by truth tables
	$\equiv$	T	by the domination law

Equivalence Proofs

Example: Show that

- $\neg[r \vee (q \wedge (\neg r \rightarrow \neg p))] \equiv \neg r \wedge (p \vee \neg q)$

Solution:

p	q	r	$\neg p$	$\neg r$	$\neg r \rightarrow \neg p$	$q \wedge (\neg r \rightarrow \neg p)$	$r \vee (q \wedge (\neg r \rightarrow \neg p))$	$\neg(r \vee (q \wedge (\neg r \rightarrow \neg p)))$	$p \vee \neg q$	$\neg r \wedge (p \vee \neg q)$
T	T	T	F	F	T	T	T	F	T	F
T	T	F	F	T	F	F	F	T	T	T
T	F	T	F	F	T	F	T	F	T	F
T	F	F	F	T	F	F	F	T	T	T
F	T	T	T	F	T	T	T	F	F	F
F	T	F	T	T	T	T	T	F	F	F
F	F	T	T	F	T	F	T	F	T	F
F	F	F	T	T	T	F	F	T	T	T

Equivalence Proofs

- Show that $[p \wedge (p \rightarrow q)] \rightarrow q$ is a tautology.

Solution:

- We use $\equiv$ to show that $[p \wedge (p \rightarrow q)] \rightarrow q \equiv T$.

$$[p \wedge (p \rightarrow q)] \rightarrow q$$

$$\longrightarrow \equiv [p \wedge (\neg p \vee q)] \rightarrow q \quad \text{substitution for } \rightarrow$$

$$\longrightarrow \equiv [(p \wedge \neg p) \vee (p \wedge q)] \rightarrow q \quad \text{distributive}$$

$$\longrightarrow \equiv [F \vee (p \wedge q)] \rightarrow q \quad \text{uniqueness}$$

$$\longrightarrow \equiv (p \wedge q) \rightarrow q \quad \text{identity}$$

$$\longrightarrow \equiv \neg(p \wedge q) \vee q \quad \text{substitution for } \rightarrow$$

$$\longrightarrow \equiv (\neg p \vee \neg q) \vee q \quad \text{DeMorgan's}$$

$$\longrightarrow \equiv \neg p \vee (\neg q \vee q) \quad \text{associative}$$

$$\longrightarrow \equiv \neg p \vee T \quad \text{excluded middle}$$

$$\longrightarrow \equiv T \quad \text{domination}$$

Equivalence Proofs

Check using a symbolic derivation whether

$$(p \wedge \neg q) \rightarrow (p \oplus r) \equiv \neg p \vee q \vee \neg r.$$

$$\begin{aligned} & (p \wedge \neg q) \Rightarrow (p \oplus r) \\ & \equiv \neg(p \wedge \neg q) \vee (p \oplus r) && [\text{Expand definition of } \rightarrow] \\ & \equiv \neg(p \wedge \neg q) \vee ((p \vee r) \wedge \neg(p \wedge r)) && [\text{Defn. of } \oplus] \\ & \equiv (\neg p \vee q) \vee ((p \vee r) \wedge \neg(p \wedge r)) && [\text{DeMorgan's Law}] \\ & \equiv (q \vee \neg p) \vee ((p \vee r) \wedge \neg(p \wedge r)) && [\vee \text{ commutes}] \\ & \equiv q \vee (\neg p \vee ((p \vee r) \wedge \neg(p \wedge r))) && [\vee \text{ associative}] \\ & \equiv q \vee (((\neg p \vee (p \vee r)) \wedge (\neg p \vee \neg(p \wedge r)))) && [\text{distrib. } \vee \text{ over } \wedge] \end{aligned}$$

Equivalence Proofs

$$\begin{aligned} & q \vee (((\neg p \vee (p \vee r)) \wedge (\neg p \vee \neg(p \wedge r))) \\ \equiv & q \vee (((\neg p \vee p) \vee r) \wedge (\neg p \vee \neg(p \wedge r))) && [\text{assoc.}] \\ \equiv & q \vee ((\mathbf{T} \vee r) \wedge (\neg p \vee \neg(p \wedge r))) && [\text{trivial taut.}] \\ \equiv & q \vee (\mathbf{T} \wedge (\neg p \vee \neg(p \wedge r))) && [\text{domination}] \\ \equiv & q \vee (\neg p \vee \neg(p \wedge r)) && [\text{identity}] \\ \equiv & q \vee (\neg p \vee (\neg p \vee \neg r)) && [\text{DeMorgan's}] \\ \equiv & q \vee ((\neg p \vee \neg p) \vee \neg r) && [\text{Assoc.}] \\ \equiv & q \vee (\neg p \vee \neg r) && [\text{Idempotent}] \\ \equiv & (q \vee \neg p) \vee \neg r && [\text{Assoc.}] \\ \equiv & \neg p \vee q \vee \neg r && [\text{Commut.}] \\ & \text{Q.E.D.} \end{aligned}$$

Remark. Q.E.D. (*quod erat demonstrandum*)